MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION DOCUMENT
49WF POTABLE WATER TREATMENT PLANT
STORM WATER DRAINAGE DESIGN

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document will list the main design criteria and design calculations relevant to the following components of the Potable water treatment plant area of Mansourah & Massarah Gold project (MMGP):

- The Potable water treatment plant stormwater system composed of open channels and pipes
- The stormwater pipes discharging some of the watersheds to the outside of the project into the natural outfalls.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

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3. SCOPE

The design document has been prepared for Potable water treatment plant Drainage System (Drawing number MD-216-49WF-EG-CV-DED-2008) based on Overall Project Layout drawing MD-216-1000-EG-ME-PPD-1001 relevant ground terracing has been obtained from site leveling drawing and necessary rainfall intensity has been captured from relevant design document.

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4. STORMWATER SYSTEM:

The stormwater system is formed of open trapezoidal channels discharging into the natural ground. These channels are connected via concrete pipes at locations that are crossing the roads or where vehicular access is expected.

4.1 Main Design Criteria

Rainfall Analysis:

The rainfall intensities and rainfall depths values were taken from the generated IDF that was indicated in approved document no. MD-216-4400-EG-ME-CAL-1001. This IDF curve is shown in Figure 1 below:

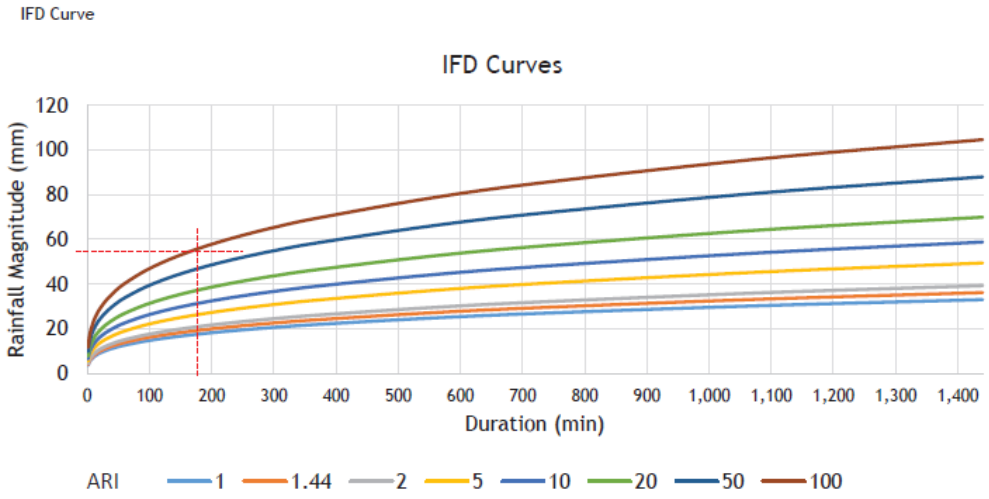


Figure 1 IDF Curve

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Table 1 below lists the derived rainfall intensity values for the 10-year return period that will be adopted in the design.

Table 1 Rainfall Intensity Values

10-year storm	10 min	20 min	30 min	60 min	120 min	180 min	360 min	720 min	1440 min
I (mm/hr)	79.58	48.9	36.8	22.7	14.0	10.5	6.5	4.0	2.5

Time of Concentration:

The time of concentration of each watershed is calculated using Kirpich Formula as follows:

$$T_c = 0.0195\left(\frac{L}{S^{0.5}}\right)^{0.77}$$

Whereas:

T_C = Time of concentration in minutes

L = Length of stream, m

S = Average slope of the stream m/m

Since most of these watersheds are relatively small, all time of concentrations are below 10 minutes, so a minimum value of 10 is adopted.

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Peak Flow Calculations:

The peak flow is calculated for each watershed using Rational Method.

The rational method is recommended for catchment areas less than or equal to 81 hectares which is the case in our project. It is an empirical formula relating rainfall intensity to runoff, and yields a peak discharge. It is usually used to calculate the peak discharge from various drainage basins as follows:

$$Q = \frac{C.I.A}{360}$$

Where:

- Q is the peak discharge, m^3/s;
- I is the average rainfall intensity, mm/hr, for a duration equal to the time of concentration and for a selected design storm;
- A is the drainage area, ha.
- C is the runoff coefficient representing land use characteristics of the watershed. Since this is an industrial area, a runoff coefficient of 0.75 is assumed.

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Hydraulic Design:

The hydraulic design of the network is done using Manning's equation, as follows:

$$V = \frac{1}{n} R^{2/3} S^{0.5}$$

Where

V, is the mean velocity of flow, in m/s;

n, is the Manning's roughness coefficient for open channel flow, n is assumed to be 0.025 for earth open channels and 0.013 for concrete pipes

R, is the hydraulic radius in m; and

S, is the slope of energy grade line, or channel bottom slope, in m/m.

The capacity of the open channel is determined from the continuity equation:

$$Q = A.V$$

Where Q is the flow rate in m^3/s,

V is the velocity in m/s, and

A is the flow area in m^2.

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4.2 Calculation Results

The following tables list the hydraulic results for the different components of the stormwater network.

Table 2 below lists the results of the watershed flows and characteristics.

Table 2 Watershed Results

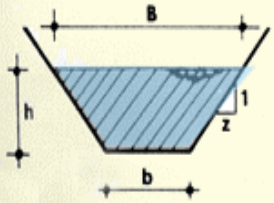
Area (A) mkd.	Drain mkd.	Length of stream (m)	Time of concentration Tc (min)	Area (ha)	Runoff coefficient `C'	Rainfall Intensity `I' (mm/hr)	Runoff from catchment (Litre/sec)	Direct input		Total Flow (Litre/sec)
								From drain mkd.	Quantity (Litre/sec)	
PWTP1	D1	30.3	10.00	0.0626	0.75	79.58	10.38			10.38
PWTP2	D2	12.6	10.00	0	0.75	79.58	0.00	D1	10.38	10.38
PWTP3	D3	85.22	10.00	0.1868	0.75	79.58	30.97	D2	10.38	41.35
PWTP4	D4	149	10.00	0.194	0.75	79.58	32.16			32.16
PWTP5	D5	103.9	10.00	0.1673	0.75	79.58	27.74			27.74
PWTP6	D6	110.6	10.00	0.1862	0.75	79.58	30.87			30.87
PWTP7	D7	9.9	10.00	0	0.75	79.58	0.00	D5 & D6	58.61	58.61

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Channel sizing

Channel Section	Wetted area	Wetted perimeter	Hydraulic radius (R)	Water top width (m)
	$(b + zh)\,h$	$b + 2h\sqrt{1+z^2}$	$\frac{(b + zh)\,h}{b + 2h\sqrt{1+z^2}}$	$b + 2zh$

Where:

b = Bottom width in m..

a = Top water width in m.

h = Depth of water in m.

z = Side slope

Table 3 Channels Hydraulic Results

Area (A) mkd.	Drain mkd.	Size of Channel							Slope	Velocity (m/sec.)	Capacity (Qcap.) (lit./sec.)
		Bottom width 'b` (m)	Depth (m)		Side slope		Top width (m)				
			Final depth (m)	Water depth 'h` (m)	H (z)	V	Top water width 'a` (m)	Top width (m)			
PWTP1	D1	0.3	0.4	0.35	1	1	1	1.1	0.0033	0.723	164.40
PWTP2	D2	Road crossing drainage pipe									
PWTP3	D3	0.3	0.625	0.575	1	1	1.45	1.55	0.002	0.731	367.74
PWTP4	D4	0.3	0.625	0.575	1	1	1.45	1.55	0.00222	0.770	387.44
PWTP5	D5	0.3	0.59	0.54	1	1	1.38	1.48	0.00285	0.843	382.58
PWTP6	D6	0.3	0.59	0.54	1	1	1.38	1.48	0.00267	0.816	370.30
PWTP7	D7	Road crossing drainage pipe									

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Table 4 below lists the results of the concrete pipes hydraulic analysis.

Table 4 Concrete Pipes Hydraulic Results

Area mkd.	Pipe Drain mkd.	Dia. of pipe`D' (m)	Slope	Velocity (m/sec.)	Capacity Qcap. (lit./sec.)	Nos. of Pipe	Qcap. (lit./sec.)
PWTP2	D2	0.3	0.005	1.10	62.38	1	62.38
PWTP7	D7	0.4	0.003	1.03	104.07	1	104.07

Considering pipe flowing at 75% full.

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1

Drainage design is covered under MD-101-SMEM-CV-CRT-0001.

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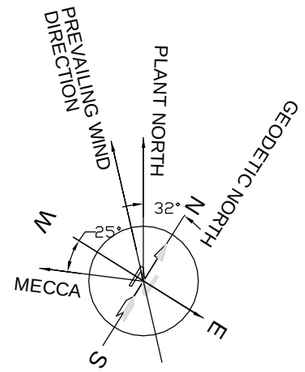
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ATTACHMENT 1 SKETCH FOR CATCHMENT AREA

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SKETCH FOR CATCHMENT AREA

PWTP1

PWTP3

PWTP5

PWTP4

PWTP6

ATTACHMENT 2 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-008676, 16 February 2021

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 16 February 2021	Comment by	EPC Response DATE: 20 February 2021	Status	Remarks
			General Comments (cc)				
MD-216-49WF-EG-CV-CAL-2007-A01	49WF	1	Drainage design is covered under MD-101-SMEM-CV-CRT-0001	Mr.Doron Golan Bechtel	Noted	Closed	